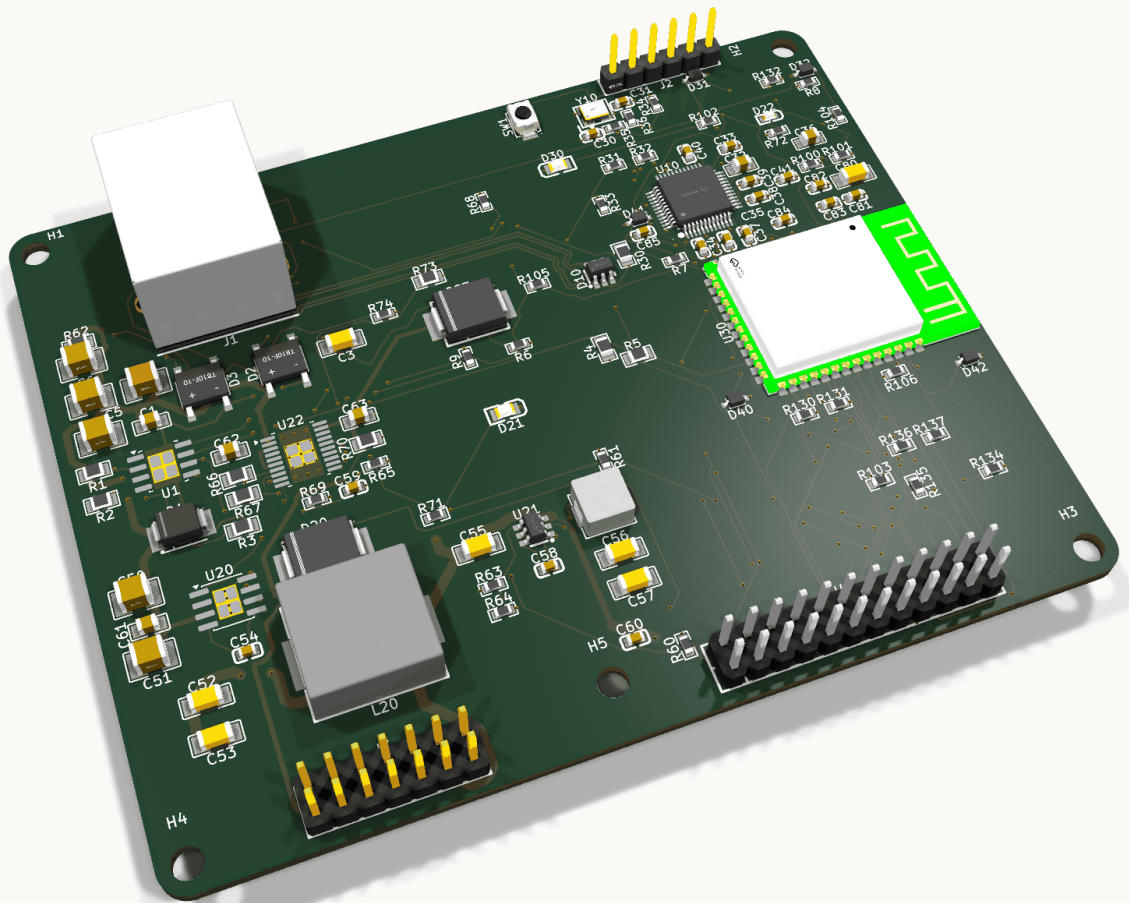


hwde

An AI PCB engineer for KiCad and JLCPCB

printed circuit boards, designed from a written brief

You describe the board you want in plain words. hwde then researches the parts, picks a topology, draws the schematic, and lays out and routes the board. It runs the electrical and manufacturing checks, and hands you the files a factory needs to build it. A human engineer signs off at every stage.

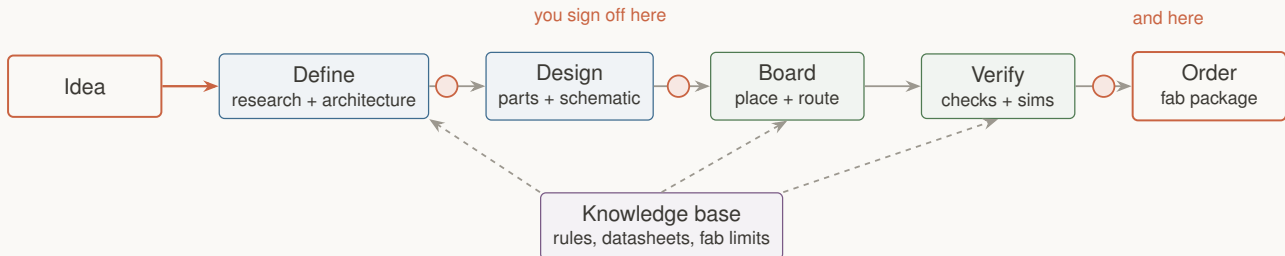


lumina-carrier is a 4-layer carrier board for a stage-lighting fixture, 100 × 80 mm with 116 footprints. hwde designed it, and it was ordered from JLCPCB and fabricated.

How it works

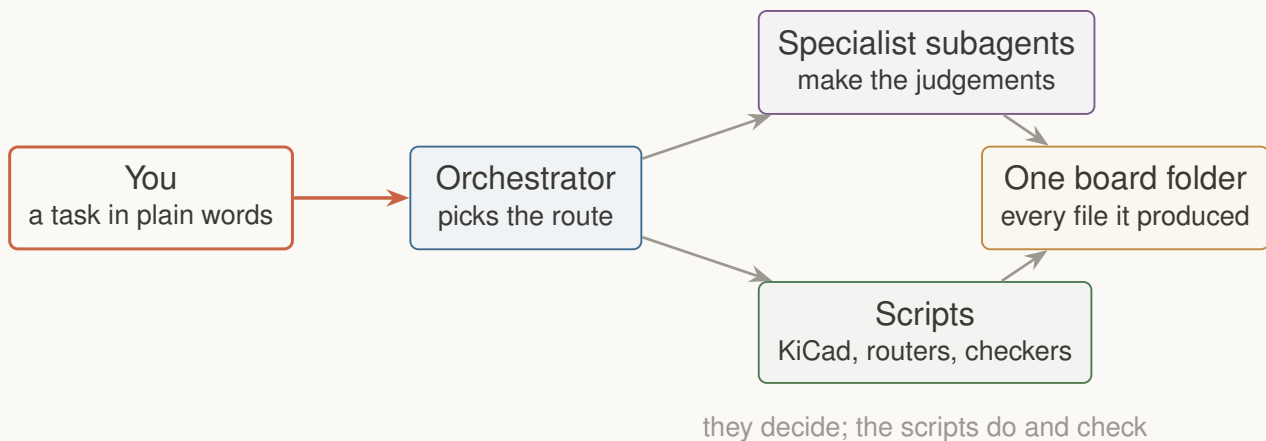
Three pictures follow. The first is the route a board takes, the second is who does which part of the work, and the third is what happens at every handover.

The pipeline



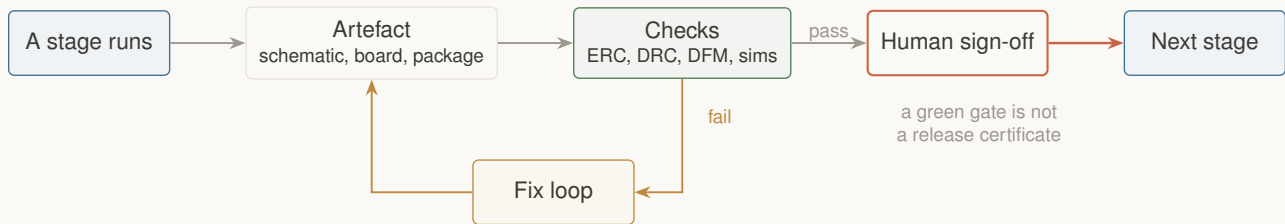
A run starts from a written brief and ends with a manufacturing package of Gerbers, a bill of materials and a pick-and-place file. Each stage in between produces one file, and before it does, it reads the knowledge base of design rules, datasheet numbers and what the factory can really make. The orange dots mark where a person has to look at the work and agree before it goes on.

Who does what



The work is split in two. Twenty-two specialist subagents make the judgement calls, like choosing a topology, sizing a part or reviewing a schematic, and each one has a written contract for what it may decide. About sixty scripts do and check the mechanical work, and they are all deterministic. They drive KiCad, run the routers and export the Gerbers. A script never has an opinion, and an agent never edits a board file by hand. Whatever either of them produces lands as a file in one folder per board.

The gate loop



Every stage ends at a gate. The gate runs the checks that apply to that stage, which are the electrical rules, the design rules, manufacturability and circuit simulations against stated bounds. It either passes, or it hands the findings back to a fix loop that reworks the file and tries again. A pass is not the end of it, because a person still signs off before the next stage starts. A green gate only proves the checks that actually ran, so the reports are kept one per check instead of being collapsed into a verdict.

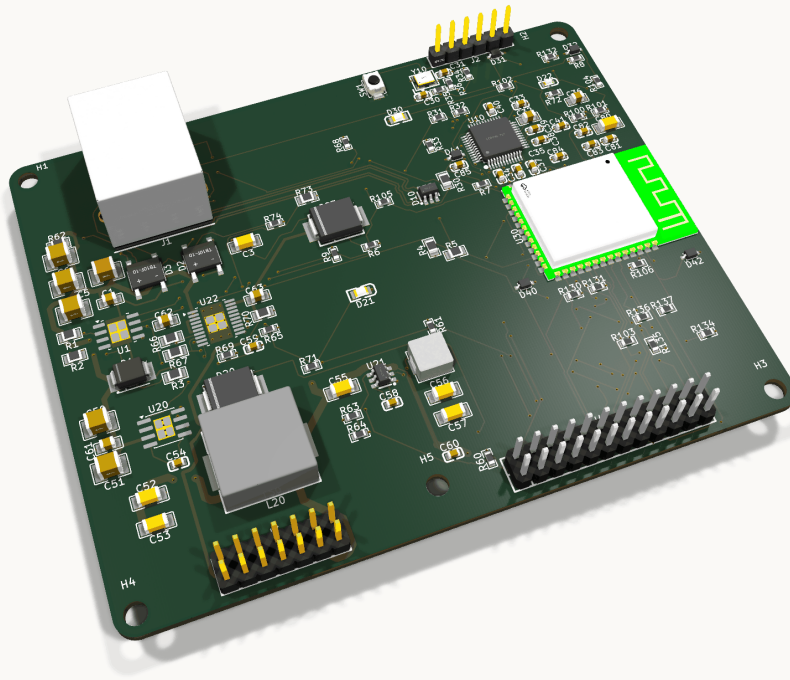
The boards

Fourteen boards got far enough to have a laid-out, routed PCB. They run from a linear regulator with five parts to a 200 W RF power stage. Every render below was made from the board file in the repository, by the script that sits next to this document.

Board	What it is	Layers	Size (mm)	Stage reached
lumina-carrier	lighting carrier, PoE-powered	4	100 × 80	ordered, fabricated
pd-trigger	USB-C PD bench trigger	2	48 × 30	ordered, fabricated
lumina-par	RGBW PAR daughter board	4	100 × 80	verification
rf-de-20m	20 MHz Class E GaN stage	4	120 × 80	manufacturability
sbuck-5v3a	5 V / 3 A synchronous buck	4	50 × 40	package ready
usb-buck	USB device dev board	4	50 × 40	package ready
g0-sense	USB-C sensor node	2	35.8 × 28.3	package ready
stm32-blinky	minimal STM32 dev board	2	50 × 40	package ready
rf-term-150w	150 W RF dummy load head	2	26 × 20	manufacturability
bb-adc	single-channel ADC	2	54.8 × 34.9	manufacturability
bb-amp	sensor amplifier chain	2	48 × 28.3	manufacturability
bb-buck	bare buck converter	2	35 × 25	manufacturability
bb-ldo	bare linear regulator	2	34.7 × 34.7	manufacturability
bb-mcu	bare microcontroller board	2	34.8 × 22.3	manufacturability

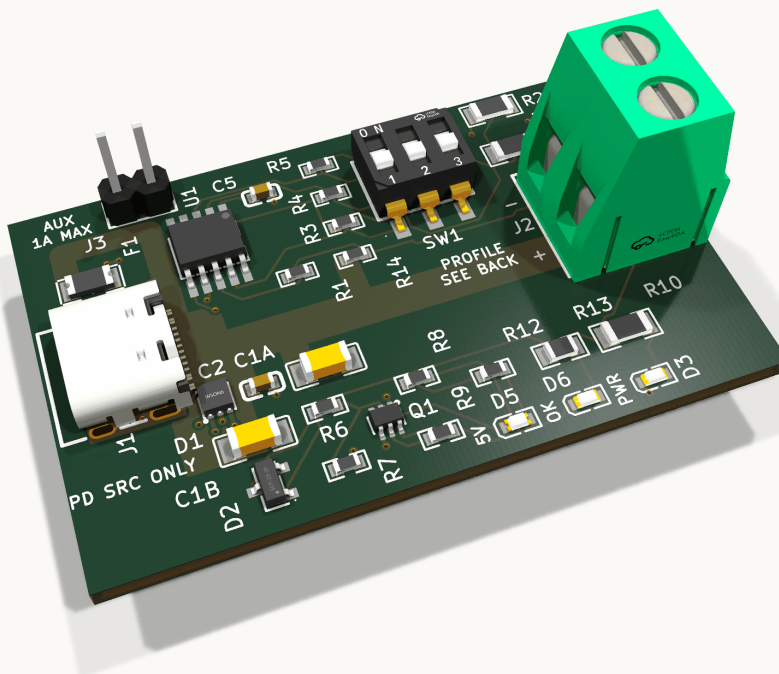
“Stage reached” says where the board stopped in the pipeline, and it does not say the board is fit to release. Two boards say “ordered, fabricated” because their own files record a JLCPCB order number that tracked through to shipped. The rest stopped with a checked manufacturing package in hand. Each size is the bounding box of the board outline, and not every outline is a rectangle.

Designed, ordered and fabricated



lumina-carrier 4 layers 100 × 80 mm 116 footprints ordered from JLCPCB, tracked to shipped

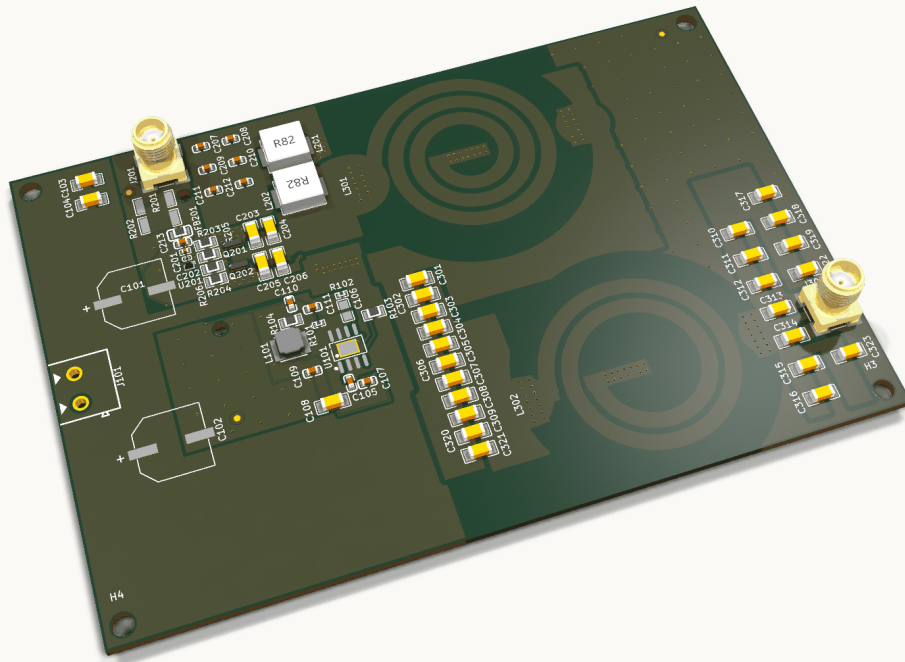
This is the universal carrier for a stage-lighting fixture. It takes power over Ethernet through a PD controller and a 100 V buck, and it carries a frozen expansion connector that a fixture-specific daughter board plugs into. It is the biggest board here, with 4 layers and 116 footprints, and another board depends on the interface it fixes.



pd-trigger 2 layers 48 × 30 mm 30 footprints ordered from JLCPCB, tracked to shipped

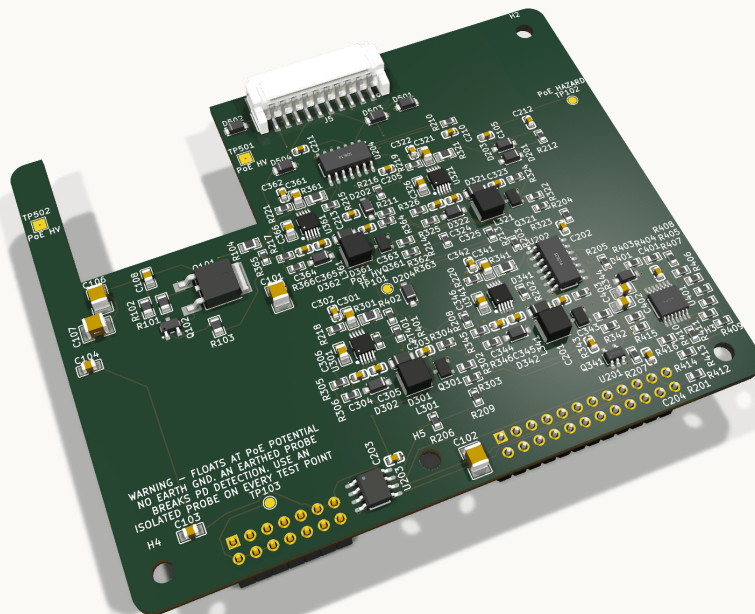
This one is a bench tool. A USB-C Power Delivery sink controller negotiates with a charger for 5, 9, 12, 15 or 20 volts, and you pick the profile on the board. The rail it wins comes out on a screw terminal rated for 5 A, so up to 100 W. The board converts nothing, it just passes the rail through.

Harder boards



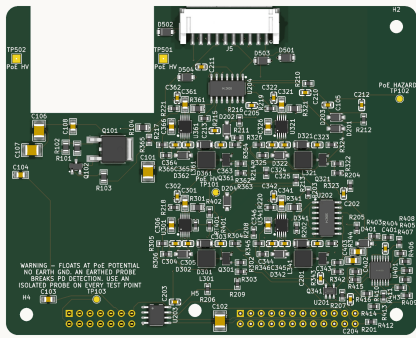
rf-de-20m 4 layers 120 × 80 mm 80 footprints stopped at manufacturability checks

A 20 MHz Class E RF power stage that puts 200 W into 50 ohms from a 40 V bus. A PWM drive arrives on an SMA input, a GaN gate driver buffers it, and one ground-referenced eGaN FET switches. A resonant tank and an L-match transform the 4.6-ohm load line to 50 ohms at the output SMA. The owner fixed the topology and the two core parts after a costed trade study. hwde did the rest.



lumina-par 4 layers 100 × 80 mm 158 footprints stopped at verification

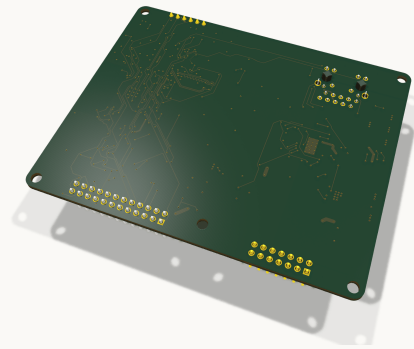
The RGBW PAR daughter board that stacks on the carrier. The carrier's connector document is frozen, so this board takes it as given and may not redefine it. Its power budget of 8.6 to 20 W was re-derived here, and depends on the power-over-Ethernet class the fixture gets. It is the densest board here at 158 footprints, and the only one whose outline is not a rectangle. The silkscreen warns that the board floats at the power-over-Ethernet potential, because an earthed probe breaks the fixture's power negotiation.



lumina-par, straight down

top-down view

The same board from straight above, so the routing shows. It has four layers and 158 footprints, routed to the same rules as everything else.

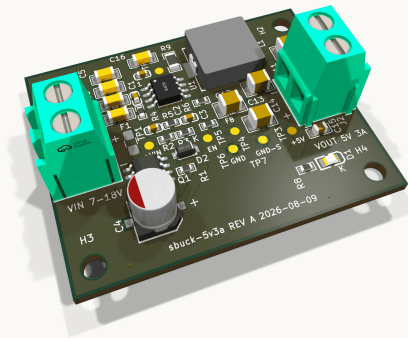


lumina-carrier, underneath

bottom view

Every part sits on the top side of this board, so the back carries routing and vias and nothing else. This is how the factory sees it.

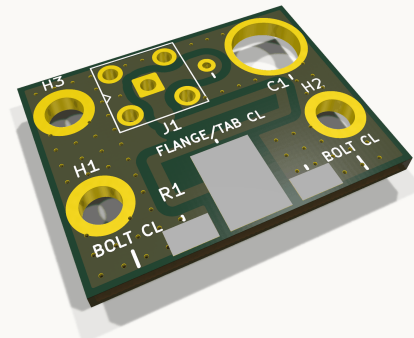
Power



sbuck-5v3a

4 layers 50 × 40 mm 44 footprints package ready

A synchronous buck that takes 7–18 V in and holds 5.0 V $\pm$ 2% at 3 A out. It is an open-frame power module.

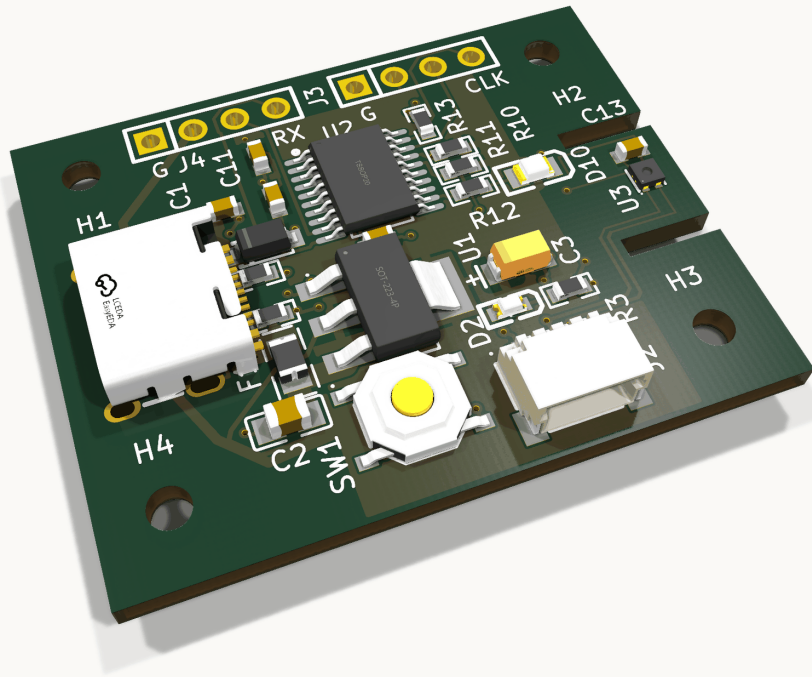


rf-term-150w

2 layers 26 × 20 mm 6 footprints stopped at manufacturability checks

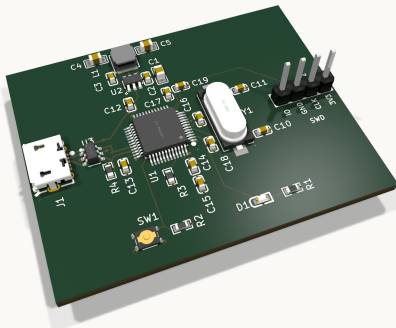
A 50-ohm, 150 W dummy load head for DC to 25 MHz. The element bolts to a heatsink the user supplies, so the board is only the RF launch and the mechanical interface. It renders close to bare because one of its six footprints has a 3D model in the repository, which leaves the connector and the element undrawn.

Microcontroller boards



g0-sense 2 layers 35.8 × 28.3 mm 30 footprints package ready, release attestation built

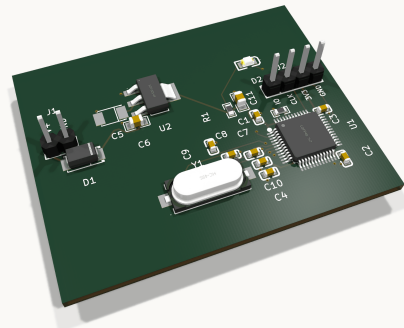
A temperature and humidity node that runs off USB-C. USB-C supplies 5 V and an LDO makes 3.3 V from it, and an STM32G030 on its internal oscillator reads a Sensirion SHT4x over I²C. The same bus comes out on a Qwiic connector, the readings go out over a UART header, and you program it over SWD. hwde designed this one end to end in a single unattended container run.



usb-buck

4 layers 50 × 40 mm 28 footprints package ready

An STM32F103 board that talks USB at full speed. The Micro-B socket carries both power and data, and there is an LED, a button and an SWD header.



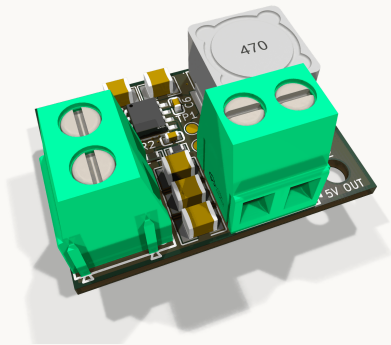
stm32-blinky

2 layers 50 × 40 mm 20 footprints package ready

The simplest useful board here. It has an STM32F103, an 8 MHz crystal, one LED, an LDO and an SWD header.

Building blocks

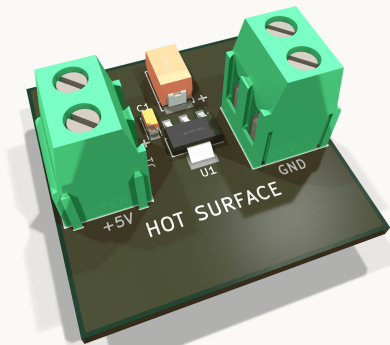
These five boards are deliberately bare, each one carrying a single block and nothing else. They are there to be studied and measured on a bench, so they leave out protection, filtering, indicators and spare rails on purpose.



bb-buck

2 layers 35 × 25 mm 20 footprints

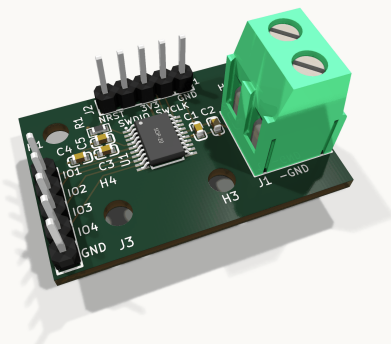
It takes 18–30 V in and gives 5 V at 2 A out on a screw terminal. It carries the converter and exactly what its datasheet requires.



bb-ldo

2 layers 34.7 × 34.7 mm 5 footprints

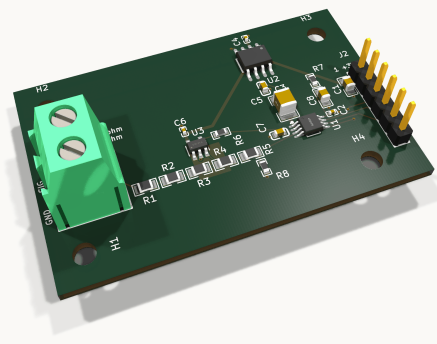
It takes 5 V in and gives 3.3 V at 500 mA out, holding regulation in still air on the board's own copper.



bb-mcu

2 layers 34.8 × 22.3 mm 14 footprints

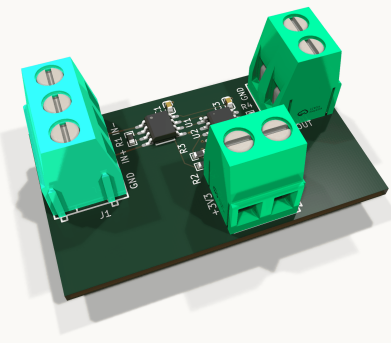
One MCU with a 3.3 V rail coming in, its debug header and four GPIO. There is no regulator and no second rail.



bb-adc

2 layers 54.8 × 34.9 mm 25 footprints

0–5 V in on a screw terminal, and 12 bits or better at up to 10 kSa/s out over a 0.1-inch header. It trades speed for DC accuracy.



bb-amp

2 layers 48 × 28.3 mm 14 footprints

It takes a 0–20 mV differential bridge signal and gives 0–3.3 V single-ended out, from DC to 1 kHz. The nominal gain is 165 V/V.

Two more boards stopped before layout and have no PCB to show, a 5 V/3 A buck and a strobe daughter board. Both are in the repository at the stage they reached.

What it can and cannot do yet

hwde is an engineering assistant, and it does not release a board by itself. A human engineer drives it and signs off at every stage. Two of the boards above were designed with it, ordered from a factory and fabricated.

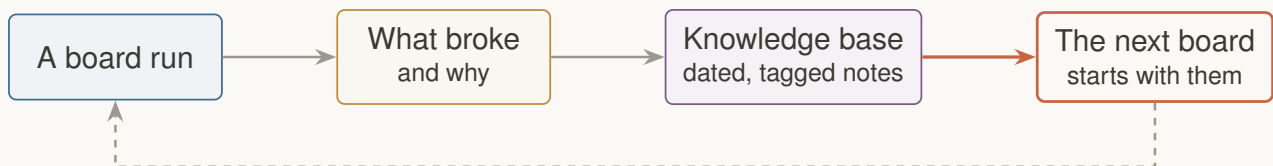
What holds

- Fourteen boards went from a written brief to a routed layout, including a 4-layer 200 W RF stage and a novel 100 W USB-C PD design.
- Two of them were ordered from a factory and fabricated, and their own files record the order numbers tracked through to shipped.
- Every check is a script, and they all follow the same contract. The findings are kept per check rather than as a single verdict.
- Everything a run produces is a file in that board's folder, and you can read it without hwde.

What does not

- A green gate only proves the checks that *ran*. Read the report for each check rather than the verdict.
- The stage a board reached is not a release certificate. A person decides what is fit to release.
- The factory has no public API for its manufacturability review, so someone still does that step by hand in a browser.
- Spending money is deliberately hard. One code path can place an order, and it handles 2-layer boards only. It refuses a board whose files already record an order, and it will not run without a fresh quote and a matching design hash. A person also has to type out a confirmation naming the board, the quantity and the total.
- The electrical and thermal checks are engineering screens, and each states its own accuracy. None of them is a certification, and nothing here is safety-certified for mains, medical, automotive or aerospace use.

How it gets better



Every run writes down what broke and why, with a date and a tag, and that goes into the knowledge base the next run reads first. The rule is to record a gotcha the moment you hit it, so the next board does not pay for it again.

The renders come from the board files in this repository, drawn by KiCad's command-line renderer. The dimensions, layer counts and footprint counts are read from those same files, and the stage each board reached is read from its recorded state.